

Algebraic Topology

HHH

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Chapter I

Homotopy

§1 Homotopy and homotopy equivalence

Definition 1.1. Given a topological space X ,

1. A **path** in X is a continuous map $\gamma : I \rightarrow X$ where $I = [0, 1]$ is the unit interval.
2. The points $\gamma(0)$ and $\gamma(1)$ are called the **initial point** and **terminal point** of the path, respectively. And $\gamma(0)$ and $\gamma(1)$ are also called the **endpoints** of the path.
3. A **loop** is a path γ such that $\gamma(0) = \gamma(1)$.

Definition 1.2. Two continuous maps $f, g : X \rightarrow Y$ are said to be **homotopic** if there exists a continuous map $H : X \times I \rightarrow Y$ such that $H(x, 0) = f(x)$ and $H(x, 1) = g(x)$ for all $x \in X$. The map H is called a **homotopy** between f and g , and we write $f \simeq g$.

Remark. The relation of homotopy on paths with fixed endpoints in any space is an equivalence relation.

Definition 1.3. The set of all homotopy classes of loops at the base point x_0 , denoted by

$$\pi_1(X, x_0)$$

is a group with respect to the product $[f][g] := [f * g]$. It is called the **fundamental group** of X at x_0 .

Proposition 1.4. If there is a path h from x_0 to x_1 , then there is an isomorphism $\pi_1(X, x_0) \rightarrow \pi_1(X, x_1)$ given by conjugation with $[f] \mapsto [h]^{-1}[f][h]$.

Corollary 1.5. If X is path-connected, then the fundamental group $\pi_1(X, x_0)$ is independent of the base point x_0 up to isomorphism.

§1.1

Proposition 1.6. *The fundamental group $\pi_1 : \mathbf{hTop}_* \rightarrow \mathbf{Grp}$ is a covariant functor from the homotopy category of pointed topological spaces to the category of groups.*

Corollary 1.7. *If X retracts onto a subspace A , then $i_* : \pi_1(A, x_0) \rightarrow \pi_1(X, x_0)$ is injective.*

Chapter II

Homology

§1

Definition 1.1. Given a sufficiently large N and consider the N -dimensional Euclidean space $\mathbb{R}^N$ and $v_0, v_1, \dots, v_n \in \mathbb{R}^N$.

1. A **n -simplex** is the convex hull of $n + 1$ points $v_0, v_1, \dots, v_n$ in $\mathbb{R}^N$ that do not lie in any $(n - 1)$ -dimensional hyperplane. It is denoted by

$$[v_0, v_1, \dots, v_n] := \left\{ \sum_{i=0}^n t_i v_i \mid t_i \geq 0, \sum t_i = 1 \right\}$$

v_i are called the **vertices** of the simplex.

2. A **subsimplex** of $[v_0, v_1, \dots, v_n]$ is a simplex of the form

$$[v_{i_0}, v_{i_1}, \dots, v_{i_k}], \quad 0 \leq i_0 < i_1 < \dots < i_k \leq n$$

For any $k < n$ and k -subsimplex, there exists canonical linear homeomorphism

$$[v_0, v_1, \dots, v_k] \rightarrow [v_{i_0}, v_{i_1}, \dots, v_{i_k}], \quad t^j v_j \mapsto t^j v_{i_j}$$

3. The subsimplex $[v_0, \dots, \hat{v}_i, \dots, v_n]$ is called the **i -th face** of $[v_0, v_1, \dots, v_n]$.

Remark. Without loss of generality, we can assume that $v_0 = 0$ and $v_i = e_i$ for $i = 1, \dots, n$, where e_i is the standard basis of $\mathbb{R}^n$. In this case, we denote the simplex by Δ^n and its i -th face by $\delta_i : \Delta^{n-1} \rightarrow \Delta^n$.

Definition 1.2. A Δ -complex structure on a space X is a collection of maps $\sigma_\alpha : \Delta^n \rightarrow X$ with n depending on α , such that

- (i) Each restriction $\sigma_\alpha|_{\text{Int } \Delta^n}$ is injective and $X = \bigsqcup_\alpha \sigma_\alpha(\text{Int } \Delta^n)$

(ii) Each restriction of σ to a face of Δ^n is equal to some σ_β , that is, for each i , there exists β such that $\sigma_\alpha \circ d^i = \sigma_\beta$.

(iii) A set $A \subset X$ is open if and only if $\sigma_\alpha^{-1}(A)$ is open in Δ^n for each α .

§2

Definition 2.1. A CW complex is a space X together with a filtration

$$\emptyset = X^{-1} \subset X^0 \subset X^1 \subset \cdots \subset X^n \subset \cdots \subset X$$

such that for each $n \geq 0$, there is a pushout diagram

$$\begin{array}{ccc} \coprod_{\alpha \in \mathcal{A}_n} S_\alpha^{n-1} & \longrightarrow & X^{n-1} \\ \downarrow & & \downarrow \\ \coprod_{\alpha \in \mathcal{A}_n} D_\alpha^n & \longrightarrow & X^n \end{array}$$

A CW-complex is finite-dimensional if $X^n = X$ for some n ; of finite type if each $\mathcal{A}_n$ is finite; and finite if it is finite-dimensional and of finite type.

§3

Definition 3.1. A **singular n -simplex** in X is a continuous map $\sigma : \Delta^n \rightarrow X$. The set of all singular n -simplices in X is denoted by $\text{Sin}_n(X)$. The **boundary operator** $\partial_n : \text{Sin}_n(X) \rightarrow \text{Sin}_{n-1}(X)$ is defined by

$$\partial_n(\sigma) := \sum_{i=0}^n (-1)^i \sigma \circ \delta_i$$

Definition 3.2. The abelian group of **singular n -chains** in X is the free abelian group $C_n(X) := \mathbb{Z}\text{Sin}_n(X)$. The complex of singular chains is the sequence of abelian groups

$$\cdots \xrightarrow{\partial_{n+1}} C_n(X) \xrightarrow{\partial_n} C_{n-1}(X) \xrightarrow{\partial_{n-1}} \cdots \xrightarrow{\partial_1} C_0(X) \xrightarrow{\partial_0} 0$$

The n -dimensional boundary $B_n(X) := \text{Im}(\partial_{n+1})$ and the n -dimensional cycle $Z_n(X) := \ker(\partial_n)$ are subgroups of $C_n(X)$. The n -th singular homology group of X is defined as the quotient group $H_n(X) := Z_n(X)/B_n(X)$.

The singular simplex, singular chain, and singular homology are functorial. That is,

$$\begin{array}{ccc}
 \mathbf{Top} & & \mathbf{Ab} \\
 \\
 \begin{array}{cccc}
 X & \text{Sin}(X) & C_*(X) & H_*(X) \\
 \downarrow f & \downarrow & \downarrow & \downarrow \\
 Y & \text{Sin}(Y) & C_*(Y) & H_*(Y)
 \end{array}
 \end{array}$$

where the map $f : X \rightarrow Y$ induces the maps.

$$\begin{array}{ccccc}
 \text{Sin}_n(X) & \xrightarrow{\partial_n} & \text{Sin}_{n-1}(X) & & C_n(X) & \xrightarrow{\partial_n} & C_{n-1}(X) & & H_n(X) & \xrightarrow{\partial_n} & H_{n-1}(X) \\
 f \circ - \downarrow & & \downarrow f \circ - & \mapsto & f_{\#} \downarrow & & \downarrow f_{\#} & \mapsto & f_* \downarrow & & \downarrow f_* \\
 \text{Sin}_n(Y) & \xrightarrow{\partial_n} & \text{Sin}_{n-1}(Y) & & C_n(Y) & \xrightarrow{\partial_n} & C_{n-1}(Y) & & H_n(Y) & \xrightarrow{\partial_n} & H_{n-1}(Y)
 \end{array}$$

In particular, Sin_n , C_n and H_n are also functors from **Top** to **Ab** or **Set**. And the natural transformations between H_n and H_{n-1} is the boundary operators ∂_n .

Theorem 3.3. *If two maps $f, g : X \rightarrow Y$ are homotopic, then they induce the homotopic chain maps, and same homomorphism.*

Proposition 3.4. *If decomposition $X = \coprod_{\alpha} X_{\alpha}$ is a disjoint union of open subsets (especially its path components), then $H_n(X) \cong \prod_{\alpha} H_n(X_{\alpha})$ for each $n \geq 0$.*

Proposition 3.5. *If X is nonempty and path-connected, then $H_0(X) \cong \mathbb{Z}$. Hence for any space X , $H_0(X)$ is a free abelian group.*

§4 Axioms of Singular Homology Theory

Let X be a topological space and $A \subset X$ be a subspace. The pair (X, A) is called a **topological space pair**. A map of space pairs $f : (X, A) \rightarrow (Y, B)$ is a continuous map $f : X \rightarrow Y$ such that $f(A) \subset B$. A continuous map $f : X \rightarrow Y$ can be viewed as a map of pairs $f : (X, \emptyset) \rightarrow (Y, \emptyset)$. This is the category of space pair **Top**² whose objects are topological pairs and morphisms are maps of space pairs. The maps $f, g : (X, A) \rightarrow (Y, B)$ are said to be **homotopic** if there is a homotopy $H : X \times I \rightarrow Y$ such that $H(A \times I) \subset B$ and $H(x, 0) = f(x)$, $H(x, 1) = g(x)$ for all $x \in X$.

For every n , there is a abelian group

$$H_n(X, A)$$

defined as $H_n(X)/H_n(A)$ (The notation $H_n(X)$ denotes $H_n(X, \emptyset)$). The continuous map $f :$

$(X, A) \rightarrow (Y, B)$ induces a homomorphism

$$f_* : H_n(X, A) \rightarrow H_n(Y, B)$$

for each n and satisfies the functoriality. In other words, we define functor

$$H_n : \mathbf{Top}^2 \rightarrow \mathbf{Ab}$$

for each n .

For each n , there is a natural transformation ∂ such that the following diagram commutes

$$\begin{array}{ccc} H_n(X, A) & \xrightarrow{\partial} & H_{n-1}(A) \\ f_* \downarrow & & \downarrow f_* \\ H_n(Y, B) & \xrightarrow{\partial} & H_{n-1}(B) \end{array}$$

Theorem 4.1 (Long exact sequence). *Given a space pair (X, A) , there is a exact sequence of chain complexes $0 \rightarrow C_*(A) \xrightarrow{i} C_*(X) \xrightarrow{j} C_*(X, A) \rightarrow 0$ where $C_n(X, A) := C_n(X)/C_n(A)$. Thus there is a long exact sequence of homology groups*

$$\cdots \rightarrow H_n(A) \xrightarrow{i_*} H_n(X) \xrightarrow{j_*} H_n(X, A) \xrightarrow{\partial} H_{n-1}(A) \rightarrow \cdots$$

Proposition 4.2. *If $f, g : (X, A) \rightarrow (Y, B)$ are homotopic maps, then they induce the same homomorphism on homology: $f_* = g_*$.*

Theorem 4.3 (Excision). *Given subspaces $Z \subset A \subset X$ such that $\overline{Z} \subset \text{Int}(A)$, then the inclusion $(X - Z, A - Z) \hookrightarrow (X, A)$ induces an isomorphism $H_n(X - Z, A - Z) \cong H_n(X, A)$ for all n .*

§5 The universal coefficient theorem

Theorem 5.1.

$$0 \longrightarrow H_n(X) \otimes G \longrightarrow H_n(X; G) \longrightarrow \text{Tor}(H_{n-1}(X), G) \longrightarrow 0$$

§6 Mayer-Vietoris sequence

Theorem 6.1. *If $X = A \cup B$ where the interiors of A and B cover X , then there is a long exact sequence*

$$\cdots \longrightarrow H_n(A \cap B) \longrightarrow H_n(A) \oplus H_n(B) \longrightarrow H_n(X) \longrightarrow H_{n-1}(A \cap B) \longrightarrow \cdots$$

§7 Kunneth formula

Theorem 7.1 (Eilenberg-Zilber).

$$C_*(X \times Y) \cong C_*(X) \otimes C_*(Y)$$

is a chain homotopy equivalence.

Theorem 7.2 (Kunneth formula).

$$0 \longrightarrow \bigoplus_{p+q=n} H_p(X) \otimes H_q(Y) \longrightarrow H_n(X \times Y) \longrightarrow \bigoplus_{p+q=n-1} \operatorname{Tor}(H_p(X), H_q(Y)) \longrightarrow 0$$

Chapter III

Cohomology

§1 Singular cohomology

Definition 1.1. Consider the R -coefficient chain complex $C_n(X; R)$, the R -coefficient cochain complex is defined as $C^n(X; R) := \text{Hom}_R(C_n(X; R), R)$

$$C_{n+1} \xrightarrow{\partial} C_n \xrightarrow{\partial} C_{n-1}$$

$$C^{n+1} \xleftarrow{\delta} C^n \xleftarrow{\delta} C^{n-1}$$

The singular cohomology group is defined as

$$H^n(X; R) := \text{Ker}()$$

Theorem 1.2.

§1.1 Cup product and cohomology ring

Definition 1.3. For cochains $\phi \in C^p(X; R)$ and $\psi \in C^q(X; R)$, the **cup product** $\phi \smile \psi$ is defined as

$$\phi \smile \psi(\sigma) := \phi(\sigma|_{[v_0, \dots, v_p]}) \cdot \psi(\sigma|_{[v_p, \dots, v_{p+q}]})$$

Lemma 1.4. For cochains $\phi \in C^p(X; R)$, $\psi \in C^q(X; R)$ and $\varphi \in C^r(X; R)$, the following properties hold:

1. Bilinearity
2. Associativity: $(\phi \smile \psi) \smile \varphi = \phi \smile (\psi \smile \varphi)$
3. The identity element: $1 \in C^0(X; R)$ is the identity element for the cup product.

Definition 1.5. The *cohomology ring* $H^*(X; R)$ is the graded ring defined as

$$H^*(X; R) := \bigoplus_{n=0}^{\infty} H^n(X; R)$$

with the cup product as the multiplication.

Remark. It has a natural R -module structure.

Definition 1.6. The *cross product or external cup product*

$$\times : H^p(X; R) \times H^q(Y; R) \longrightarrow H^{p+q}(X \times Y; R)$$

is defined as

$$\phi \times \psi := p_1^* \phi \smile p_2^* \psi$$

where $p_1 : X \times Y \rightarrow X$ and $p_2 : X \times Y \rightarrow Y$ are the projection maps.

Proposition 1.7. 1. $\delta(\phi \smile \psi) = \delta\phi \smile \psi + (-1)^p \phi \smile \delta\psi$

$$2. \delta(\phi \smile \psi) = (-1)^{p+q} \delta(\psi \smile \phi)$$

$$3. \delta \text{ has naturity: } f^*(\phi \smile \psi) = f^*\phi \smile f^*\psi$$